

Kubernetes Deployment YAML — Beginner Friendly + Detailed Explanation

■ YAML with Inline Comments

```
apiVersion: apps/v1 # API version for Deployment resources
kind: Deployment # Type of resource (Deployment)
```

```
metadata: # Information about the Deployment
name: nginx-deployment # Name of the Deployment
namespace: dev # Namespace where it will be created
labels: # Labels for identifying the Deployment
app: nginx # Key-value label
```

```
spec: # Desired state of the Deployment
replicas: 3 # Number of Pods to run
```

```
selector: # How Deployment finds its Pods
matchLabels:
app: nginx # Must match Pod labels
```

```
template: # Blueprint for Pods
metadata:
labels:
app: nginx # Labels applied to Pods
```

```
spec: # Pod specification
containers: # List of containers in the Pod
- name: nginx # Container name
image: nginx:1.24 # Container image
ports:
- containerPort: 80 # Port exposed by container
```

■ Simple Explanation (Beginner Friendly)

This YAML creates:

- A **Deployment** named `nginx-deployment`
- Inside namespace `dev`
- That runs **3 Pods**
- Each Pod runs **Nginx (version 1.24)**

■ Block-by-Block Explanation

1. API + Kind

- `apiVersion: apps/v1` → Uses Deployment API
- `kind: Deployment` → We are creating a Deployment

2. Metadata

- `name` → Name of Deployment
- `namespace` → Where it is created
- `labels` → Used for grouping and selection

3. Spec (Desired State)

- `replicas: 3` → Keep 3 Pods running

4. Selector

- Tells Deployment which Pods belong to it
- Must match Pod labels exactly

5. Template (Pod Blueprint)

- Defines how Pods should look
- Includes labels + container config

6. Pod Spec

- Defines containers inside Pod

Container Details:

- Name: nginx
- Image: nginx:1.24
- Port: 80

■ How It Works Internally

When you run:

```
kubectl apply -f deployment.yaml
```

Kubernetes will:

1. Create Deployment
2. Deployment creates ReplicaSet
3. ReplicaSet creates 3 Pods
4. Each Pod runs nginx container

■ Flow Diagram

Deployment

↓

ReplicaSet

↓

Pods (3)

↓

Containers (nginx)

■■ Important Rules

Label Matching

These MUST match:

```
selector.matchLabels.app = nginx
```

```
template.metadata.labels.app = nginx
```

Otherwise Deployment will break ■

■ Final Summary

- Deployment manages Pods automatically
- Keeps desired number of replicas running
- Uses labels to track Pods
- Pods run containers

■ One-Line Summary

■ This YAML creates a Deployment that runs **3** nginx Pods in the dev namespace.